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Generative AI use in EFL writing: associations with originality, critical reasoning, and metacognitive engagement in a Turkish higher education context

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ABSTRACT

Generative artificial intelligence (AI) writing tools are increasingly embedded in university students' writing practices, yet little is known about how their use relates to higher-order writing skills in English as a Foreign Language (EFL) contexts. This mixed-methods study explored associations between generative AI use and the originality, critical reasoning, and metacognitive engagement of first-year students enrolled in a public university in Türkiye. Seventy-eight undergraduates completed three timed, in-class argumentative essays with optional access to ChatGPT (idea generation), Grammarly (language suggestions), and QuillBot (paraphrasing). Data included rubric-based ratings of the essays, Torrance Tests of Creative Thinking (originality subscale), Cognitive Reflection Test scores, reflective journals, and focus-group interviews. Findings revealed two distinct usage patterns. Students who frequently relied on AI tools produced linguistically fluent and well-organised texts but displayed lower originality and less developed argumentation, often reporting cognitive offloading and diminished authorial control. In contrast, students who used AI selectively, mainly for language refinement or initial brainstorming, maintained a stronger personal voice, demonstrated greater critical reasoning, and engaged more actively in planning and reflection. These results emphasise the need for explicit guidance in AI supported writing instruction: although AI can reduce lower-level linguistic strain, uncritical dependence may limit opportunities for cognitive effort and meaningful learning in academic writing.

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Generative AI writing tools; self-regulated learning; cognitive load theory; sociocultural theory; EFL writing

1. Introduction

Generative AI writing assistants, such as ChatGPT, Grammarly, and QuillBot have become increasingly embedded in students' academic literacy practices. Recent international surveys indicate that a large majority of university students routinely experiment with AI tools during coursework, often using them to overcome linguistic challenges, generate ideas, or refine written products. This rapid integration

is particularly salient in English as a Foreign Language (EFL) settings and higher education contexts, where students simultaneously navigate language demands and complex academic writing conventions. While AI tools may alleviate lower-level burdens such as grammar and vocabulary, concerns have emerged regarding their potential influence on higher-order writing processes, including originality, critical thinking, and metacognitive engagement. The pedagogical question for higher education is therefore not only whether AI can support writing, but how students' use of AI reshapes their cognitive and reflective involvement in the writing process.

Existing research provides valuable but incomplete insights. Many studies focus predominantly on linguistic outcomes—accuracy, fluency, cohesion, and error reduction—showing generally positive effects when students use AI tools for surface-level improvement. Far less is known about how AI mediates deeper dimensions of writing, such as idea generation, argument development, or students' capacity to regulate and monitor their writing choices. Prior work also tends to conceptualise AI tool use as either clearly beneficial or clearly detrimental, offering limited empirical evidence on the nuanced ways students actively appropriate or over-rely on AI. In addition, despite theoretical discussions about cognitive offloading, authorship, and identity, empirical investigations into how students perceive and reflect on their use of AI remain scarce, especially in non-Western higher education contexts. This gap is consequential, as EFL writers often experience distinct linguistic pressures and may interact with AI tools differently from L1 writers.

The present study responds to these gaps by examining patterns of AI use rather than their causal effects and by framing AI-assisted writing as a mediated cognitive, metacognitive, and sociocultural activity. Drawing on Cognitive Load Theory (CLT), we consider how AI may redistribute mental effort during writing. Through the lens of Self-Regulated Learning (SRL), we explore how students plan, monitor, and evaluate their writing when AI is available. From a Sociocultural Theory (SCT) perspective, we interpret AI as a mediating artefact that may support or disrupt students' developing academic voice and authorial identity. Together, these perspectives provide an integrated framework for understanding the complex ways students engage with AI during writing.

Guided by this framework, the study adopts a non-causal, exploratory orientation to investigate how EFL students use, interpret, and reflect on AI tools during academic writing in a higher education context. Specifically, the study addresses the following research questions:

RQ1. What patterns of AI tool use emerge among EFL students during academic writing, and how are these patterns associated with originality, critical reasoning, and metacognitive engagement?

RQ2. How do EFL students perceive and reflect on their AI use at different stages of the writing process?

By focusing on patterns of interaction rather than treatment effects, the study aims to provide theoretically grounded, context-sensitive insight into how generative AI is reshaping EFL academic writing and the cognitive and reflective processes that underpin it.

2. Theoretical framework

The study draws on an integrated theoretical framework combining Cognitive Load Theory (CLT), Self-Regulated Learning (SRL), and Sociocultural Theory (SCT) to interpret how students use (rather than are affected by) generative AI tools during academic writing. These theories illuminate complementary dimensions of students' engagement with AI—cognitive processes, regulatory behaviours, and sociocultural mediation—and together provide a coherent lens for examining patterns of originality, critical thinking, and metacognitive engagement.

2.1. Cognitive Load Theory (CLT)

CLT (Sweller, 1988, 2023) posits that learning outcomes depend on the distribution of intrinsic, extraneous, and germane cognitive load. In L2 writing, students must simultaneously manage linguistic complexity, organisation, idea generation, and argumentation, all of which contribute to cognitive load. Generative AI tools may alter this distribution: proofreading tools can reduce extraneous load, while idea-generation tools may simplify aspects of intrinsic load by offering conceptual scaffolds.

However, CLT also highlights that excessive reliance on external aids may reduce germane load, meaning students engage less in the deep processing necessary for learning—such as generating original ideas or constructing arguments. In this study, we explore how students perceive and manage these load dynamics when AI tools are available. Rather than assuming AI produces specific effects, we examine how different usage patterns correspond to different forms of cognitive engagement, particularly in originality and critical reasoning.

2.2. Self-regulated learning (SRL)

SRL theory conceptualises writing as a process involving planning, strategy selection, monitoring, and reflection (Oxford, 2011; Zimmerman & Risemberg, 1997). Generative AI tools introduce new decision points into this cycle: students must determine when, why, and how to use AI, evaluate AI output, and integrate or reject suggestions.

Students with stronger SRL skills may treat AI as a strategic aid, using it in ways aligned with their goals (e.g. checking language after drafting or using suggestions as prompts for deeper thinking). In contrast, students with weaker SRL tendencies may adopt a more reactive approach, turning to AI whenever uncertain and potentially bypassing essential phases of the writing process.

In this study, SRL serves as a framework for examining students' metacognitive engagement with AI—how they regulate their writing behaviour, how they monitor the AI's contributions, and how they reflect on its role in their learning.

2.3. Sociocultural theory (SCT)

SCT views learning as mediated by cultural tools and social interaction (Vygotsky, 1978). Generative AI introduces a new mediational artefact into the writing activity

system. AI tools may function as a form of scaffolding within students' zones of proximal development, offering linguistic formulations, examples, or organisational templates. However, SCT also emphasises voice appropriation and the development of authorial identity. When AI-generated language is incorporated uncritically, students may experience voice dissonance, where parts of the text feel disconnected from their own authorial stance (Yin & He, 2023).

In this study, SCT helps interpret how students negotiate authorship, ownership of ideas, and the social meaning of AI use—especially in relation to originality and perceived academic integrity.

2.4. Integrated theoretical perspective

Rather than treating these theories separately, we adopt an integrated perspective. CLT explains cognitive patterns in how students allocate effort; SRL accounts for the strategic and metacognitive choices underlying those patterns; SCT illuminates how students interact with AI tools as cultural artefacts and how those interactions shape writing identity. Together, these lenses guide our examination of how students use AI tools, how they make sense of their choices, and how different usage patterns correspond to differences in originality, critical thinking, and metacognitive engagement.

3. Literature review

3.1. Cognitive dimensions of AI writing assistance

The integration of AI powered writing tools into EFL higher education has generated growing scholarly attention to their cognitive implications, particularly regarding how such tools may scaffold or suppress deeper thinking processes. Several studies indicate that AI support can reduce cognitive burden by assisting with lower-level linguistic choices. For instance, Yang and Rui (2025) found that AI enhanced learning environments can increase EFL students' engagement by alleviating emotional barriers such as anxiety and depression, which may indirectly reduce the cognitive strain associated with language learning. In a related vein, Liu et al. (2021) similarly found that AI-prompted revision nudged students toward greater self-monitoring and reflection—processes that align with core tenets of SRL, such as metacognitive regulation. Within the lens of CLT, these findings suggest that AI tools may reduce extraneous cognitive load by automating surface-level decisions, allowing learners to allocate more cognitive resources to content development and argumentation.

However, the same affordances may unintentionally encourage cognitive offloading. Barrett and Pack (2023) warned that students frequently accept AI suggestions without evaluating their accuracy or relevance, potentially bypassing critical reflection and diminishing cognitive effort. Recent experimental work by Zhang et al. (2025) similarly suggests that students who rely heavily on generative feedback may engage less in higher-order revision. These studies collectively suggest that AI's cognitive impact is not uniformly beneficial; rather, it depends on how strategically students engage with the assistance they receive. This dual potential underscores the

importance of examining student–tool interaction patterns, metacognitive awareness, and decision-making strategies in AI supported writing.

3.2. Creative aspects of AI-supported writing

Creativity in second-language writing involves the ability to generate novel, context-appropriate ideas under genre and linguistic constraints. Generative AI has shown potential to support divergent thinking, especially in brainstorming, lexical exploration, and reframing of arguments. Guo and Li (2024) demonstrated that tailored chatbots facilitated richer ideation and improved student confidence during initial drafting. Marzuki et al. (2023) likewise noted stylistic experimentation and lexical variation among Indonesian EFL learners producing essays with AI suggestion features.

Yet concerns emerge regarding authenticity and voice. Werdiningsih et al. (2024) found that frequent AI support made some students feel detached from their writing identity, perceiving AI-generated phrasing as overly formulaic. Yin and He (2023) argued that generative systems often converge toward conventional discourse patterns, potentially narrowing rather than expanding creative exploration. These findings highlight the need to conceptualise creativity not only as fluent idea generation, which tools may enhance—but also as authorship and self-expression, which tools may undermine if students are not encouraged to critically adapt suggestions. In this regard, SCT offers a critical lens: interaction with tools mediates participation in the writing community, shaping learners’ voice and agency. Recent work highlights that AI-mediated environments—such as conversational bots and immersive virtual spaces—are reshaping how learners participate in language learning communities and develop digital literacy (Godwin-Jones, 2023).

3.3. Student perceptions and engagement with AI tools

Learners’ perceptions strongly influence how they regulate AI use and benefit from its scaffolds. Multi-country survey research indicates that EFL students often approach AI assistance with cautious optimism. In Indonesia, Fithriani (2023) reported that students appreciated reduced anxiety and conceptual clarification, yet worried about skill loss and plagiarism. Comparable ambivalence has been documented in Algeria (Guendouz & Benettayeb, 2025), where students valued efficiency but expressed fears of overdependence and diminished authorship.

Contextual factors also shape these perceptions. In Türkiye, Tümen Akyıldız (2024) observed that students lacked guidance on how to meaningfully integrate feedback from AI tools into their revision plans, resulting in inconsistent strategy use. In North American settings, Rudolph et al. (2023) found that students saw AI as a helpful collaborator but felt uneasy when generated phrasing did not match their personal writing voice (Griffith & Schultz, 2022). Together, these studies suggest a global convergence: students recognise AI’s potential but require clearer instructional frameworks to navigate ethical considerations, identity formation, and strategic use.

Despite this growing body of research, little is known about how L1 mediation influences AI assisted writing behaviours in bilingual settings. Preliminary

observations suggest that some EFL learners may generate ideas in their first language (L1) and rely on AI to translate or rewrite them, potentially affecting voice development and cognitive processing (Rudolph et al., 2023). This dimension remains largely understudied and is particularly relevant to the Turkish higher education context examined in the present study.

3.4. Synthesis and identified gaps

Collectively, the literature illustrates that AI writing tools can scaffold both cognitive and creative demands by automating linguistic tasks and supporting ideation. At the same time, concerns persist regarding dependency, reduced authorship, and diminished engagement with cognitively complex aspects of writing. A nuanced understanding of AI's role thus requires attention not only to performance outcomes but also to how students regulate and interpret tool-mediated writing experiences—a key intersection of CLT, SRL, and SCT.

Three major gaps remain. First, despite theoretical claims about cognitive and creative impact, relatively few studies adopt robust mixed-methods designs that triangulate quantitative performance metrics with qualitative measures of learner experience. Second, while research spans diverse regions, there is limited work in Turkish higher education contexts, where L1–L2 interaction patterns and sociocultural attitudes toward technology may uniquely shape AI engagement. Third, AI use is often examined at a single point in time rather than longitudinally, leaving open questions about sustained influence on learners' voice, authorship, and strategic self-regulation.

The current study addresses these gaps by conducting a theory-informed, exploratory mixed-methods investigation into how Turkish EFL undergraduates experience AI-assisted academic writing over a semester. By integrating performance measures with reflection-based accounts, this research seeks to move beyond binary judgments of AI as purely advantageous or harmful, instead foregrounding the complexities of learner–tool interaction.

4. Methodology

4.1. Research design

This study adopted a convergent parallel mixed-methods design (Creswell & Plano Clark, 2018), in which quantitative and qualitative strands were implemented during the same academic term, analysed separately, and then integrated during interpretation. The design was chosen to allow an exploratory examination of how naturally occurring patterns of AI tool use were associated with EFL students' originality, critical thinking, and metacognitive engagement in academic writing, without manipulating AI use as an experimental treatment or making causal claims.

The study was conducted over one 10-week semester in a first-year Academic Writing course. Quantitative data comprised students' performance on three timed in-class argumentative essays and standardised measures administered at multiple time points. Qualitative data consisted of reflective journals and focus group

interviews that captured students' evolving perceptions, strategies, and self-reported engagement with AI tools. Treating both strands as equally important, we used the qualitative data to illuminate the processes behind the quantitative patterns and to refine our interpretation of the associations observed.

4.2. Participants and context

Participants were 78 first-year undergraduates (approximately 18–19 years old) enrolled in a compulsory Academic Writing course at a large public university in Türkiye. All were native speakers of Turkish writing in English as an L2, with intermediate proficiency (B1–B2 on the CEFR), as determined by a university placement test administered prior to the course. Students represented a range of majors (e.g. engineering, education, social sciences) but shared a similar status as novice academic writers in English, making them an appropriate cohort for examining how generative AI tools are appropriated in early university writing.

Ethical approval was obtained from the university's Institutional Review Board, and all students provided informed consent. The study was embedded in normal course activities to minimise disruption. Students were informed that participation was voluntary, that their choice to use or not use AI tools would not affect course grades, and that all data (essays, logs, and reflections) would be anonymised and used solely for research purposes.

The Academic Writing course focused on argumentative essay writing, including formulating a thesis, supporting claims with evidence, organising ideas coherently, and maintaining an academic tone. Typically, students drafted essays, received instructor feedback, and revised their work. In this study, we augmented the existing curriculum by making AI tools available as optional supports during specific in-class writing tasks. The course remained an in-person, instructor-led environment; AI tools were positioned as supplementary resources rather than replacements for instruction.

4.3. AI tool integration and writing tasks

Three generative and AI-enhanced writing tools were integrated to represent distinct types of support:

ChatGPT (OpenAI) was introduced as a conversational assistant that could generate ideas, outline arguments, and propose example content in response to prompts.

Grammarly was presented as an AI enhanced writing assistant focusing on grammar, vocabulary, tone, and style suggestions, primarily for local-level language feedback.

QuillBot was described as a paraphrasing and rewriting tool designed to help students explore alternative phrasings and improve sentence fluency.

4.3.1. Training and guidelines

During the first two weeks of the course, students participated in two 60-minute tutorial sessions on responsible and strategic AI use. These tutorials, led by the

course instructor, combined live demonstrations with guided practice. For each tool, we modelled (a) how to formulate prompts (e.g. asking ChatGPT for an outline rather than a full essay), (b) how to interpret and critically evaluate AI suggestions, and (c) how to revise AI-generated language so that it aligned with the student's intended meaning and voice.

Explicit guidelines on academic integrity were discussed and shared in writing. Students were told that AI tools should support their thinking and writing (e.g. as brainstorming partners or language checkers) rather than replace it. Concrete examples were provided of acceptable versus problematic uses, and we emphasised that undisclosed wholesale copying of AI generated text would violate expectations for original work. These preparatory sessions aimed to scaffold students' metacognitive awareness of when and how to draw on AI tools during writing.

4.3.2. In-class writing tasks and AI access

Over the semester, students completed three major in-class argumentative essays (approximately in Weeks 3, 5, and 7). Each task required a 250–300-word essay written in 60 min in a computer lab on contemporary topics relevant to students' lives (e.g. “Does technology make university students learn more effectively?”; “Should university instruction be in English or the national language?”).

During each writing session, students had access only to the three designated AI tools and the word-processing environment; other internet browsing was blocked. Students could decide whether to use any of the tools and, if so, at which stages (planning, drafting, revising) and for which purposes (e.g. generating examples, checking grammar, paraphrasing). The instructor and a teaching assistant monitored the lab to ensure that students adhered to the guidelines (e.g. no pasting of entire AI-generated essays).

Immediately after each task, students completed a brief AI Usage Log, indicating (a) which tools they used, (b) at which stages of the writing process, and (c) for what functions (e.g. “used ChatGPT to brainstorm arguments,” “used Grammarly to check grammar during revision,” “did not use AI”). Lab observations (e.g. visible copy-paste behaviour from ChatGPT into the essay editor) were also noted. These data were later used to construct AI use profiles for exploratory comparisons (see [Section 4.5](#)).

4.4. Instruments

4.4.1. Rubric-based evaluation of essays

All essays were evaluated using an analytic rubric developed for this study, informed by established frameworks for L2 writing assessment (e.g. Green, 2014; Lee, 2017; Zimmerman & Risemberg, 1997). The rubric contained four 5-point scales:

Argumentation and Critical Thinking—clarity of thesis, quality and coherence of reasoning, and use of relevant evidence or examples.

Organisation and Coherence—effectiveness of overall structure (introduction, body, conclusion), logical sequencing of ideas, and use of transitions.

Creativity and Originality—presence of insightful or non-formulaic ideas, originality of examples or perspectives, and avoidance of overly generic responses.

Voice and Authorship—consistency of tone and phrasing that reflects the student's own voice, with lower scores indicating impersonal, inconsistent, or AI-dominated language.

The rubric was iteratively refined. A preliminary version was piloted on a set of 10 essays from a previous cohort not included in the study, after which descriptors and anchor examples were clarified. Two experienced EFL writing instructors, blind to students' identities and AI use profiles, independently rated all essays. Inter-rater reliability was high (Cohen's $\kappa = .88$ across dimensions), and disagreements were resolved through discussion. Average scores across raters provided quantitative indicators of each student's performance on the four dimensions.

4.4.2. Torrance tests of creative thinking (TTCT)

To capture students' general divergent thinking ability, we administered the TTCT Verbal—Short Form in Week 1 (pre-course) and Week 10 (end of course). The TTCT-Verbal includes tasks such as generating unusual questions or consequences for hypothetical situations. Following the test manual, responses were scored for fluency (number of relevant ideas), originality (statistical infrequency of ideas), and elaboration (level of detail). In this study, we focused on the Originality subscore as an index of creative ideation, acknowledging that it does not directly measure academic writing creativity but provides a complementary, domain-general indicator of students' capacity to generate novel ideas.

4.4.3. Cognitive reflection test (CRT)

The three-item Cognitive Reflection Test (Frederick, 2005) was administered four times: at the beginning of the course (pre-test) and immediately after each of the three essay tasks. The CRT presents problems with an intuitively appealing but incorrect answer; correct responses require suppressing intuition and engaging in analytic reasoning. Scores ranged from 0 to 3 correct. In this study, CRT performance was used as a proxy for students' tendency to engage in reflective, "System 2" thinking (Kahneman, 2011) rather than to accept immediate answers—a disposition that is theoretically relevant to how learners may interact with AI suggestions.

4.4.4. Reflective journals

To examine metacognitive engagement and students' evolving experiences with AI, each student maintained a reflective journal throughout the course. After each major writing task (Weeks 3, 5, and 7), students wrote a 150–200-word reflection in response to prompts about how they approached the task, whether and how they used AI tools, and how they felt about the resulting essay (e.g. "Did the essay feel like it was written in your own voice?"). In Week 10, they composed a 300–400-word summative reflection on their overall experience with AI in the course and perceived changes in their writing process.

Reflective journals are widely used to capture metacognitive awareness and self-regulated learning processes in writing and higher education (Boud & Falchikov, 2007; Chen et al., 2021; Zimmerman & Risemberg, 1997). In this study, they provided rich, time-sensitive accounts of students' planning, monitoring, and evaluation of their AI use.

4.4.5. Focus group interviews

After the final writing task, we conducted two semi-structured focus group interviews, each with six volunteer students. Participants were purposively sampled to include a mix of frequent, moderate, and minimal AI users based on their logs and observed behaviour. The 45-min sessions explored topics such as perceived benefits and drawbacks of AI tools, changes in attitudes over the semester, and strategies for integrating AI into writing. The interviews were audio-recorded with consent and transcribed verbatim.

4.5. Data analysis

4.5.1. Quantitative analysis

Using AI usage logs and observations, we constructed AI use profiles for each student across the three writing tasks. For exploratory analysis, we focused on students located at the clearly differentiated ends of the observed usage continuum, avoiding arbitrary cut-offs and ensuring that the contrasted profiles represented meaningfully distinct engagement patterns rather than minor variations in tool use.

Frequent AI users—students who reported and were observed using at least one of the AI tools in all or most tasks and for multiple functions (e.g. idea generation and language editing).

Minimal AI users—students who reported little or no AI use across tasks, composing largely without AI assistance.

A small number of students with intermediate usage patterns were retained in the dataset but not emphasised in between-group contrasts. Because AI use was self-selected, these groupings are treated as descriptive profiles rather than experimental conditions.

For each student, analytic rubric scores were averaged across the three essays for each dimension (Argumentation, Organisation, Creativity, Voice). We then conducted exploratory independent-samples t-tests comparing frequent and minimal users on these averaged scores, on TTCT Originality gain scores (post-test minus pre-test), and on CRT scores at each time point. Where appropriate, we also examined trends using a 2 (group) $\times$ 4 (time) mixed ANOVA for CRT scores. Statistical analyses were conducted using SPSS, with alpha set at .05. Effect sizes (Cohen's d and partial η^2) were calculated to gauge the magnitude of differences. Given the non-experimental, self-selection design, these analyses are interpreted as associations rather than evidence of causal impact.

4.5.2. Qualitative analysis

Journal entries and focus group transcripts were analysed using thematic analysis (Braun & Clarke, 2006), with constructs from CLT, SRL, and SCT serving as sensitising

concepts rather than rigid categories. In the first cycle, two researchers read all texts and conducted open coding to identify salient ideas (e.g. “using AI as shortcut,” “learning from AI feedback,” “loss of personal voice,” “strategy change over time”). In the second cycle, codes were grouped into candidate themes and refined through constant comparison. Through iterative discussion, we arrived at four overarching themes: (1) surface-level reliance on AI, (2) collaborative and controlled use of AI, (3) voice and authorship issues, and (4) evolving metacognitive awareness of AI use.

To enhance trustworthiness, the two researchers independently coded a subset of journals and both focus group transcripts, then compared and reconciled discrepancies to refine the coding scheme. Representative quotations were selected to illustrate each theme. Member checking was conducted informally by summarising preliminary themes to a subset of participants and inviting feedback on whether the interpretations resonated with their experiences. Finally, qualitative themes were integrated with quantitative findings during the interpretation phase to address the research questions in a holistic and theory-informed manner.

5. Results

We present the results in relation to the two research questions. For RQ1, we describe the patterns of AI tool use that emerged in the course and how these patterns were associated with indicators of originality, critical reasoning, and metacognitive engagement. For RQ2, we report students’ perceptions of AI tools and their reflections on how AI shaped their writing processes. Quantitative and qualitative findings are reported together within each subsection where they speak to the same aspect of the research questions.

5.1. RQ1: Patterns of AI use and associations with originality, critical reasoning, and metacognitive engagement

5.1.1. AI use profiles

Based on AI usage logs and lab observations, two contrasting AI use profiles were identified at the ends of a continuum: frequent AI users and minimal AI users. Frequent users reported drawing on at least one of the three tools (ChatGPT, Grammarly, QuillBot) in all or most of the timed essays, often at multiple stages (planning, drafting, revising). Minimal users reported little or no AI use across the three tasks, typically composing their essays without AI assistance or using AI only for occasional language checks.

These profiles were exploratory and descriptive rather than experimentally manipulated groups. A small number of students with moderate or inconsistent use patterns fell between these two profiles and are not the focus of the between-group contrasts reported below, although their qualitative accounts informed theme development.

5.1.2. Quantitative associations between AI use profiles and performance indicators

Table 1 summarises descriptive and inferential statistics comparing frequent and minimal AI users on key quantitative indicators. For each student, rubric scores for

Table 1. Key performance differences by AI usage group (Minimal vs. Frequent users).

Measure (metric)	Minimal AI users (M)	Frequent AI users (M)	Group difference
TTCT Originality (post-test score, 0–100)	80	70	Minimal > Frequent ($p = .008$)
Cognitive Reflection Test (score, 0–3)	2.7	2.3	Descriptive trend: Minimal > Frequent (n.s.)
Organisation & Coherence (rubric, 1–5)	4.0	4.5	Frequent > Minimal ($p < .001$)
Argumentation & Reasoning (rubric, 1–5)	4.2	3.8	Minimal > Frequent ($p = .042$)
Creativity & Originality (rubric, 1–5)	4.0	3.6	Minimal > Frequent ($p = .037$)
Voice & Authorship (rubric, 1–5)	4.1	3.5	Descriptive difference—not significance-tested

Note. Higher scores indicate better performance on all measures. TTCT: Torrance Tests of Creative Thinking (Verbal, Originality subscale); CRT: Cognitive Reflection Test. p = exact significance values. Because AI use was self-selected, all comparisons are descriptive in nature and do not imply causal effects.

the three essays were averaged on four dimensions: Organisation & Coherence, Argumentation & Reasoning, Creativity & Originality, and Voice & Authorship. In addition, TTCT Originality post-test scores and CRT scores were examined.

Writing rubric scores. On average, frequent AI users obtained higher scores on Organisation & Coherence than minimal users ($M=4.5$ vs. 4.0 on a 5-point scale; $p < .001$), indicating more consistently well-structured essays with clearer textual organisation. In contrast, minimal AI users obtained higher mean scores on Argumentation & Reasoning (4.2 vs. 3.8 ; $p = .042$), Creativity & Originality (4.0 vs. 3.6 ; $p = .037$), and Voice & Authorship (4.1 vs. 3.5 ; descriptive difference—not significance-tested), suggesting that their essays were judged to contain more original ideas and to reflect a more consistent personal voice.

TTCT originality. Minimal AI users showed higher mean scores on the TTCT Originality subscale at post-test ($= 80/100$) than frequent users ($= 70/100$), with this difference reaching statistical significance ($p = .008$). Within the limitations of the non-experimental design, this pattern indicates that, in this sample, students who reported minimal reliance on AI tended to show larger gains in divergent-thinking originality over the semester.

CRT performance. On the Cognitive Reflection Test, both groups improved slightly over time. By the final administration, minimal AI users obtained a mean score of 2.7 out of 3 , whereas frequent users averaged 2.3 . This difference did not reach statistical significance (n.s.), but the descriptive trend is consistent with somewhat stronger performance on this reflective-reasoning measure among minimal users.

In brief, these quantitative results suggest that frequent AI use in this cohort was associated with stronger text organisation, whereas minimal use was associated with higher ratings on originality, argumentation, and voice, as well as higher TTCT originality scores. Because AI use was self-selected, these patterns are interpreted as associations rather than causal effects.

In addition, effect sizes for the significant differences were small to moderate (Cohen’s $d = .30-.55$), indicating that while statistically reliable, the magnitude of performance differences between usage profiles was not large. This is consistent with the exploratory, self-selected nature of the design, in which tool use was not experimentally controlled.

5.1.3. Agency in AI tool use

Qualitative data provided further insight into how students exercised agency over AI tools within these profiles. Two contrasting patterns emerged: surface-level reliance and collaborative, controlled use.

5.1.3.1. Surface-level reliance. Many students classified as frequent AI users described treating AI tools as an “easy button” for writing. One student wrote, “If I got stuck, I just asked ChatGPT and whatever it replied, I kind of copy-pasted it in.” Another explained, “Grammarly fixed my sentences, so I didn’t really review that part myself.” Such accounts depict AI use as a way to complete the task efficiently, with limited attention to how suggestions were generated or how they related to the student’s own reasoning. This pattern aligns with the quantitative finding that frequent users tended to obtain higher organisation scores but lower originality and argumentation scores: they produced structurally coherent texts but often with generic content and limited evidence of deeper engagement.

5.1.3.2. Collaborative and controlled use. In contrast, students whose AI use was more selective and deliberate described AI as a helpful partner rather than a substitute. For example, one journal entry noted, “I used ChatGPT to brainstorm ideas. It gave me a list of points for and against. I didn’t use all of them, but it helped me see different angles.” Another student commented in a focus group that Grammarly functioned like “a personal proofreader,” adding, “It pointed out a lot of my mistakes, and I took that feedback and tried to understand why it was wrong so I don’t repeat it.” These students reported actively evaluating AI suggestions, rephrasing outputs to fit their intended meaning, and sometimes rejecting suggestions that did not align with their voice or the task. Their essays often fell into the minimal or moderate AI use category and were typically rated higher on originality and voice.

Across the semester, many students described a shift in their AI use. Early entries emphasised novelty (e.g. “ChatGPT is amazing—it gave me an introduction in seconds”), whereas later reflections showed more restraint: “In the first essay I relied on ChatGPT a lot and I realised I didn’t remember much of what I wrote. So in the third essay I only used it to check grammar.” Another student concluded, “I learned that I should use AI for ideas, but then do the writing myself. Otherwise I’m not practising my writing.” These accounts indicate increasing awareness of the need to regulate AI use in ways that match their learning goals.

5.1.4. Originality, voice, and associations with AI use

Qualitative evidence further illuminated the quantitative differences in originality and voice between AI use profiles. Several students in the frequent-use group reported a sense of voice dissonance, feeling that parts of their essays no longer sounded like their own writing. One wrote, “The paragraph ChatGPT wrote sounded very formal and not like how I usually write. It didn’t feel like me.” Another reflected that after heavy use of QuillBot, “the sentence was grammatically perfect but I wouldn’t normally use those words, so it felt off.” Some students connected this to

a reduced sense of ownership: “I got a high grade, but I don’t feel 100% proud because the AI basically generated the examples I used.”

By contrast, students who reported minimal or targeted AI use generally felt that their essays still reflected their own voice. Raters likewise noted that low-AI essays tended to have a more consistent personal tone, whereas some high-AI essays showed noticeable shifts in diction and style, as if written by different authors. These observations align with the rubric pattern in which minimal users’ texts were rated higher on Creativity & Originality and Voice & Authorship.

A comparison of two essays on the prompt “Should university classes be taught in English in non-English-speaking countries?” illustrates this pattern. One student, who reported minimal AI use, drew on personal experience and described both advantages and challenges of university courses taught in English, including an anecdote about a classmate who struggled until summaries were provided in Turkish. This essay received higher originality and voice ratings. Another student, classified as a frequent AI user, produced a linguistically polished and well-organised essay that emphasised common arguments (“English is a global language,” “students need English for their careers”) without specific, contextualised examples. This essay received lower originality and voice ratings despite high scores for organisation.

5.1.5. Critical reasoning and metacognitive engagement

The association between AI use patterns and critical reasoning was reflected in both rubric scores and students’ descriptions of how they handled AI suggestions. In terms of argumentation, essays from minimal AI users more often included qualified claims, explicit consideration of counterarguments, and specific examples. In contrast, essays from frequent users were sometimes described by raters as “descriptively adequate but analytically thin,” providing lists of advantages and disadvantages without deeper evaluation.

Student accounts help explain these patterns. One frequent user shared, “ChatGPT gave me an opposing viewpoint that I hadn’t thought of. I didn’t really agree with it, but I included it anyway because it made my essay look more complete.” Another student, who used AI more selectively, reported a different approach: “It gave me a counterargument, but I actually disagreed with it. So I wrote my own rebuttal to that idea and just used Grammarly to polish the phrasing.” These contrasting accounts illustrate how similar AI prompts can be integrated into essays in ways that either bypass or foreground students’ own critical reasoning.

With regard to metacognitive engagement, students described a spectrum of planning and monitoring behaviours. Some reported drafting their essay independently and then consulting AI for local language refinement (“I plan what I want to say first, then I write a draft. If I’m unsure about wording, I ask the AI, but I don’t accept everything.”). Others, more commonly in the frequent-use profile, reported turning to AI whenever they encountered difficulty (“I ended up using whatever QuillBot suggested even if the style was different. I was just relieved it fixed my paragraph.”).

Reflections written at the end of the course suggest that many students became more aware of these patterns over time. Comments such as “I have to remain in control of the essay, otherwise I’m not really improving” and “AI is useful, but I

shouldn't depend on it too much" indicate that students were beginning to articulate criteria for regulating their own AI use.

In summary, for RQ1, the quantitative and qualitative data together indicate that different AI use profiles in this cohort were associated with distinct patterns in originality, argumentation, and perceived authorship, as well as with different approaches to planning and monitoring writing. These patterns are descriptive and do not imply that AI use itself caused particular outcomes.

5.2. RQ2: Students' perceptions and reflections on AI use

For RQ2, we distinguished between students' perceptions of AI tools (how they viewed and evaluated AI as part of their writing environment) and their reflections (how they interpreted AI's role in their learning and adjusted their strategies over time).

5.2.1. Perceptions of benefits and risks

Across journals and focus groups, students described AI tools in predominantly positive terms at the start of the course. Many characterised ChatGPT, Grammarly, and QuillBot as convenient supports that reduced anxiety, provided quick language correction, and generated ideas when they felt stuck. Typical comments included, "It gave me an introduction in seconds," "I feel more confident because Grammarly catches my mistakes," and "ChatGPT helped me think of arguments I would not think of alone."

At the same time, students expressed reservations and ambivalence. Several voiced concerns about becoming "lazy" or "dependent" on AI, about getting "too used to it," or about "cheating themselves" out of practice. Others worried about fairness ("Some students may use AI more and get higher grades") or about the boundaries of acceptable use ("I'm not sure how much AI is allowed in homework"). These perceptions mirror broader debates about AI and academic integrity and highlight the need for clearer guidance, which students themselves explicitly requested.

5.2.2. Reflections on learning and strategy change

Beyond initial perceptions, many students used their journals and the final reflection to examine how AI had shaped their own writing process over the semester. These reflections often involved evaluative statements about the quality of their learning, not just about the tools themselves.

Several students reported that heavy AI use early in the course led them to feel detached from their own writing or unsure about what they had actually learned. One wrote, "In the first essay, I relied on ChatGPT a lot, and I realised I didn't remember much of what I wrote." Another observed, "Using ChatGPT made my essay structure better, but I think I became lazy to come up with arguments myself." These reflections were often followed by descriptions of deliberate changes in strategy, such as attempting a later essay without AI or limiting AI use to post-draft grammar checks.

Other students described AI as a source of feedback that contributed to their learning when used critically. For example, a student noted, “It pointed out a lot of my mistakes, and I tried to understand why it was wrong, so I don’t repeat it.” Another explained, “AI gave me ideas, but I always tried to ask: does this really make sense for my topic?” These comments suggest that some learners integrated AI feedback into their self-regulation routines by questioning suggestions and using them as prompts for deeper thinking.

Taken together, these reflections indicate that students were not only forming attitudes toward AI but were also monitoring and reassessing their own patterns of use. By the end of the course, many articulated a desire for balance, expressing intentions to “use AI for support but not for everything” and to “write more on my own first and then check with AI.” In this sense, RQ2 is addressed by showing how students both perceived AI tools (as helpful but potentially risky) and reflected on the implications of their use for their own learning, often leading to adjustments in strategy.

6. Discussion

This study explored how generative AI tools were associated with students’ academic writing processes and products in an EFL university context. The findings suggest that AI use can support or hinder higher-order writing development depending on whether students retain agency in decision-making. The discussion below interprets these associations through CLT, SRL, and SCT, highlighting how these frameworks jointly illuminate the divergent trajectories observed.

6.1. *AI as scaffold or crutch: Interpreting patterns through CLT*

Results indicate that AI tools appeared to reduce extraneous cognitive load by automating mechanical aspects of writing, such as grammar and basic structuring. This aligns with research demonstrating increased fluency and reduced error density when AI-supported revision assists with lower-level issues (e.g. Guo & Li, 2024; Marzuki et al., 2023). In this respect, frequent AI users produced well-organised texts with apparent ease.

However, CLT emphasises that efficiency may be detrimental when it displaces germane cognitive load—the effort devoted to idea generation and reasoning (Sweller, 2023). The profiles observed here suggest that some students, particularly frequent users, offloaded conceptual decisions to AI, which was associated with lower ratings for originality and argumentative depth. This pattern aligns with emerging evidence that AI-driven or hybrid feedback may bypass essential cognitive struggle (Niloy et al., 2024; Zhang et al., 2025). Thus, from a cognitive perspective, the educational challenge is not whether AI should support students, but how to ensure that support does not replace the intellectual work needed for learning.

6.2. *Self-regulation as a determinant of productive AI use*

Differences in metacognitive engagement help explain why similar AI tools corresponded to different outcomes. Students who demonstrated strategic planning,

monitoring, and evaluation tended to use AI judiciously and maintain responsibility for key aspects of writing. They treated suggestions as hypotheses to assess, consistent with SRL models where tools extend, rather than replace, learners' strategy sets (Yang et al., 2023; Zimmerman, 2002). In contrast, students who employed AI reactively—particularly when encountering difficulty—were more likely to accept suggestions uncritically, allowing the tool to direct arguments and style.

This observation parallels findings in technology-enhanced learning that self-regulation shapes whether digital tools promote or undermine learning (Barrett & Pack, 2023; Hadwin et al., 2018). The present study suggests that SRL is a moderating factor: AI supported writing is most beneficial when accompanied by explicit goal-setting, iterative monitoring, and post-use reflection.

6.3. Sociocultural mediation and voice appropriation

SCT perspectives provide further insight into how AI tools mediate the writing activity system. When AI suggestions were selectively interpreted and revoiced, students retained authorship and expressed a coherent personal stance—an indication of successful appropriation of a mediating tool. Conversely, when AI contributed major stretches of text, some students experienced voice dissonance, finding that parts of their writing no longer aligned with their linguistic identity. This challenge echoes concerns in L2 writing that generative models may homogenise style and diminish authorial agency (Fithriani, 2023; Yin & He, 2023).

Thus, from an SCT perspective, the observed patterns show that the educational impact of AI does not reside in the tool itself, but in how learners negotiate its role within their meaning-making process. Supporting students to critically integrate AI contributions rather than adopt them wholesale may help sustain identity formation and ownership in academic writing.

6.4. Positioning within current CALL and L2 writing research

The present study corroborates prior evidence that AI can improve surface-level quality while simultaneously posing risks to originality and argumentative depth (e.g. Bai & Wang, 2023; Werdiningsih et al., 2024). It extends this literature in three ways. First, rather than treating improved accuracy and reduced originality as separate phenomena, this study links both to cognitive load distribution and regulatory decisions. Second, unlike studies focusing on fully AI generated texts, this context involved AI assisted but human-authored writing, showing that dependency effects can occur even when students remain active composers. Third, students increasingly recognised the trade-offs of AI support and attempted to rebalance tool reliance, suggesting potential for AI use to catalyse metacognitive growth when appropriately guided.

Importantly, the observed associations are interpretive rather than causal, given the self-selected nature of AI use. Nonetheless, the convergent patterns across quantitative and qualitative indicators underscore the educational significance of learner agency in shaping the role of AI.

7. Implications

This study highlights several pedagogical implications for thoughtfully incorporating AI in higher education and broader EFL writing courses. These implications reflect the need for clear norms, strategic scaffolding, and support for authorship and agency.

7.1. Establish guidelines on the scope and boundaries of AI use

Students expressed uncertainty regarding acceptable levels of AI involvement. Institutions may help resolve this by developing policies that distinguish between supportive and substitutive AI functions (e.g. permissible for language refinement or brainstorming; restricted for generating entire arguments) and by emphasising transparency in reporting how AI is used (Rudolph et al., 2023). Such clarity may reduce ethical ambiguity and encourage honest reflection about tool use.

7.2. Provide explicit instruction on strategic tool use

Findings suggest that benefits are greatest when AI is incorporated into self-regulated processes. Therefore, writing instruction should include AI literacy components, such as modelling workflows where students draft first before consulting tools, teaching strategies for evaluating AI-generated content, encouraging fact-checking and critical questioning of outputs, and integrating post-use reflection on learning impact. This aligns with calls in CALL research to support not only technical competence but also metacognitive control in AI use (Liu et al., 2021; Syahnaz & Fithriani, 2023).

7.3. Foster voice development and ownership

Given the risk of voice dilution, instructors can incorporate activities that promote appropriation, such as asking students to identify and revoice AI suggested phrasing to align with their personal tone, comparing alternative formulations to examine how authorship is conveyed, and discussing what constitutes original contribution in AI-supported contexts. Such practices reflect SCT's emphasis on identity formation in mediated writing and help maintain authenticity and confidence in authorship.

7.4. Support equitable access and guidance through academic services

Writing centres and learning support units may need to expand services to include AI use consultation, enabling students to discuss strategies for balancing assistance with effort. This may be especially beneficial for learners with weaker SRL skills, helping them make informed decisions rather than relying reactively on the tool.

7.5. Reframe AI as a context for metacognitive growth

Because students in this study increasingly adapted their approaches over time, educators can view AI not simply as a challenge but as an opportunity to develop

self-regulation and academic judgement. Reflective tasks that ask how AI shaped decision-making may help students internalise productive norms.

8. Conclusion, contributions, and future directions

8.1. Conclusion

Across one university-level writing course, distinct patterns of AI tool use were associated with different learning outcomes. Students who relied heavily on AI tended to produce well-organised and grammatically accurate essays but showed less originality, weaker argumentation, and a reduced sense of authorship. Conversely, students who used AI strategically and sparingly were more likely to demonstrate engagement in idea generation, critical reasoning, and personal voice, even when their language contained more errors. These findings indicate that generative AI can lower surface-level barriers, yet its learning value depends on the extent to which students remain cognitively and metacognitively engaged in the writing process.

This study contributes empirical evidence to a nuanced position: AI in EFL academic writing is neither inherently beneficial nor detrimental. Rather, student agency—the ability to regulate, interpret, and appropriate AI outputs—plays a decisive role in determining whether AI functions as a productive scaffold or a counterproductive shortcut.

8.2. Theoretical contributions

By applying CLT, SRL, and SCT in an integrated manner, this study advances the-
orisation of AI supported writing in three ways. First, at the cognitive level, patterns observed show that AI tools can reduce extraneous load (surface corrections) but may also inadvertently reduce germane load if students outsource essential reasoning and idea generation. Second, at the regulatory level, the findings empirically reinforce that planning, monitoring, and reflection are crucial in determining whether AI is used constructively, extending SRL theory to include AI literacy as part of the strategy repertoire required for academic writing. Third, at the sociocultural level, the study highlights voice appropriation as a key site of mediation: when students fail to internalise AI generated language, voice dissonance emerges, and AI risks overshadowing the writer's identity.

Together, these findings contribute a multi-level explanation for the divergent experiences learners report when writing with AI, offering a theoretical foundation for future CALL scholarship.

8.3. Pedagogical implications for university writing contexts

The results underscore a pressing need for explicit instruction on strategic AI integration, rather than assuming students will self-regulate automatically. Practical guidance includes clearly defining task boundaries (e.g. requiring student-generated thesis statements, outlines, and arguments before any AI consultation), teaching

critical evaluation routines (e.g. checklists for assessing AI suggestions), preserving and developing voice (e.g. marking and revising AI influenced sentences in one's own tone), and supporting reflective awareness through structured reflections or micro-journals on why, when, and how AI was used.

These implications emphasise that engagement, not avoidance, is the pedagogical imperative. By modelling and supervising effective AI use, teachers can help students maintain originality and authorship while still benefiting from AI's affordances.

8.4. Policy and ethical implications

Institutional policies should go beyond binary restrictions to provide clear operational norms that protect both integrity and learning. Possible components include specifying permitted uses (e.g. language editing, idea brainstorming), restricted uses (e.g. full-paragraph generation without substantial revision or attribution), simple disclosure practices (e.g. a brief statement on how AI assisted the writing), and campus-wide AI literacy initiatives for both students and instructors.

Such policies reduce uncertainty, discourage covert reliance, and reinforce the principle that academic work must remain meaningfully the student's own. Transparency also acknowledges the evolving realities of writing in higher education, rather than pushing AI use underground.

8.5. Limitations and future research

As a self-selected usage design, the associations observed here do not support causal claims. Future work should therefore employ experimental or quasi-experimental designs (e.g. random assignment to structured AI use conditions), propensity score or profile matching to reduce self-selection bias, and longitudinal tracking to examine whether divergences widen or diminish over time.

Additional promising directions include studying discipline-specific writing (e.g. argumentative versus technical genres), investigating affective dimensions of AI-assisted writing (e.g. confidence, ownership), using screen recordings or think-aloud protocols to capture real-time AI–writer interaction, and comparing different types of AI support (feedback-first vs. content-first systems). As generative AI evolves, understanding how students adapt—not just their products but their thinking processes—will be central to maintaining educational rigour in EFL writing.

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